



CFD ANALYSIS OF COPPER AUTOMOBILE RADIATOR

STEADY-STATE THERMAL ANALYSIS

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Heat Transfer Operations

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COPPER RADIATOR CFD – GROUP 1



TEAM MEMBERS

& THEIR CONTRIBUTIONS

STEP 1 – GEOMETRY CREATION

Prakrut created and cleaned the radiator geometry in ANSYS DesignModeler to ensure a clean model.

STEP 2 – GEOMETRY INTEGRATION

The finalized geometry was imported and linked with ANSYS Fluent for further setup.

STEP 3 – MESHING

Anagha generated a tetrahedral mesh with 0.005 m element size and defined named selections for all boundaries.

STEP 4 – SETUP & POST-PROCESSING

Aarav configured materials, zones, and boundary conditions in Fluent.
Simulation and result analysis were performed collaboratively by all members.

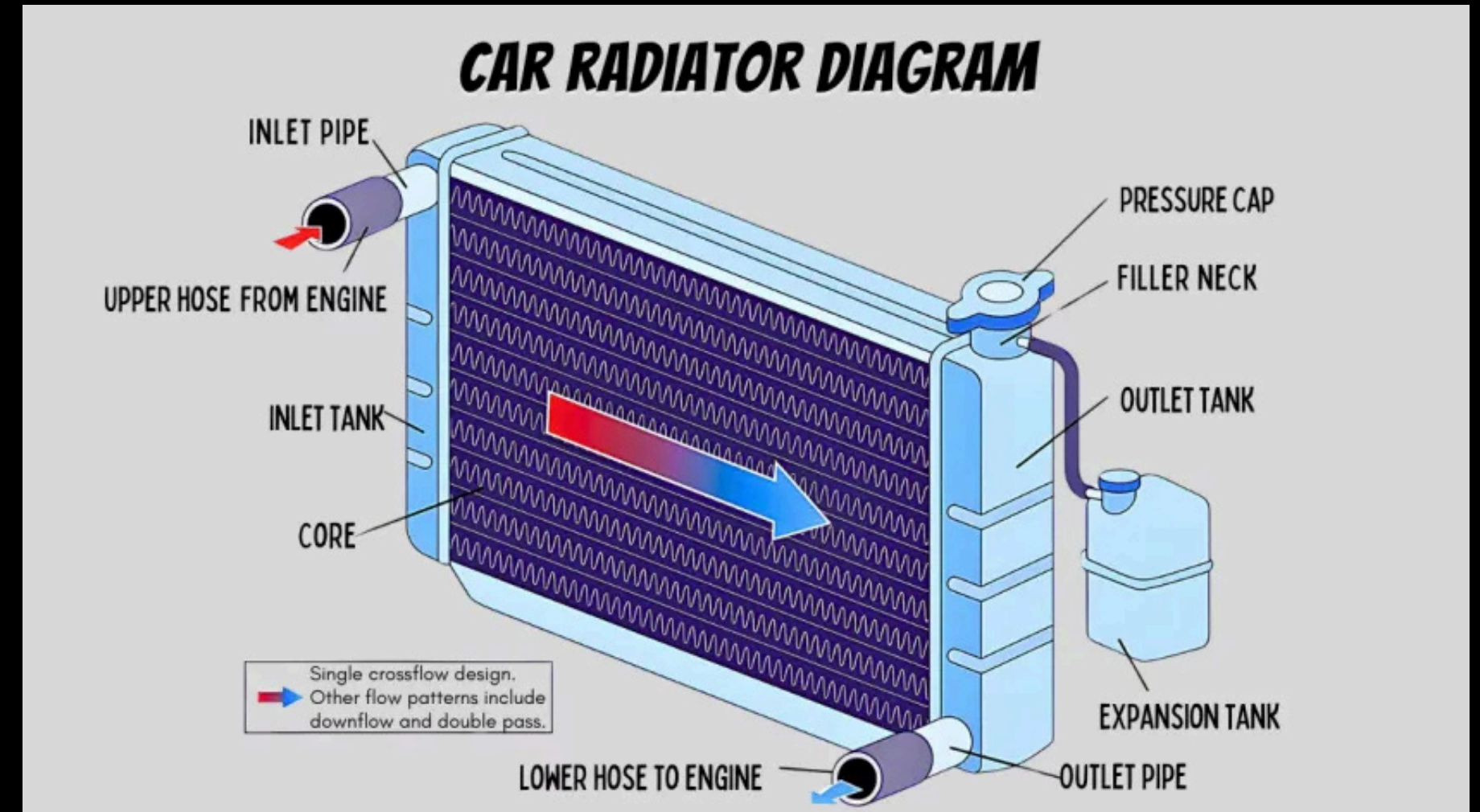
➤ INTRODUCTION

The automobile radiator is a key component for engine cooling, enabling heat exchange between the coolant and ambient air.

Objective:

- Simulate the fluid flow and heat transfer inside a radiator core .
- Evaluate temperature distribution, pressure drop, and wall heat flux performance.
- Finally, to evaluate the overall cooling performance and discuss possible design improvements.

The CFD simulation is performed in ANSYS Fluent to analyze heat transfer between the coolant (water) and the solid (copper).



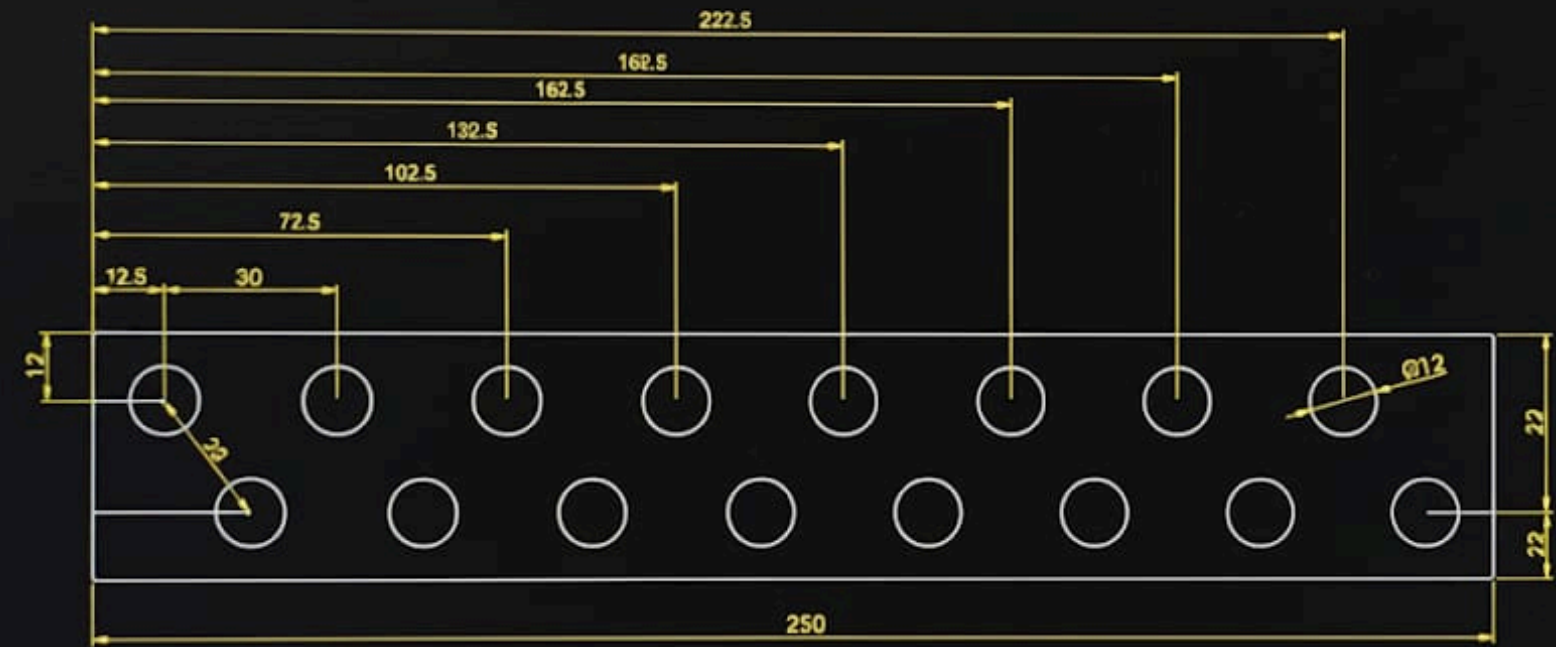
SOURCE: THE ENGINEERING CHOICE

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➤ GEOMETRY AND DOMAIN SETUP

Geometry Selection :

- Rectangular copper plate: 250mm×290mm×44mm
- 16 holes (8 per row × 2)
- Hole diameter: $\varnothing 12$ mm
- Hole spacing: 30 mm (center to center)
- Edge clearance: 12.5 mm from left, 12 mm from top/bottom
- Designed symmetrically for uniform coolant flow



➤ GEOMETRY AND DOMAIN SETUP

The geometry was created using ANSYS DesignModeler. The radiator model consists of tubes, fins, inlet/outlet sections, the fluid domain, and solid walls.

Simplifications Applied:

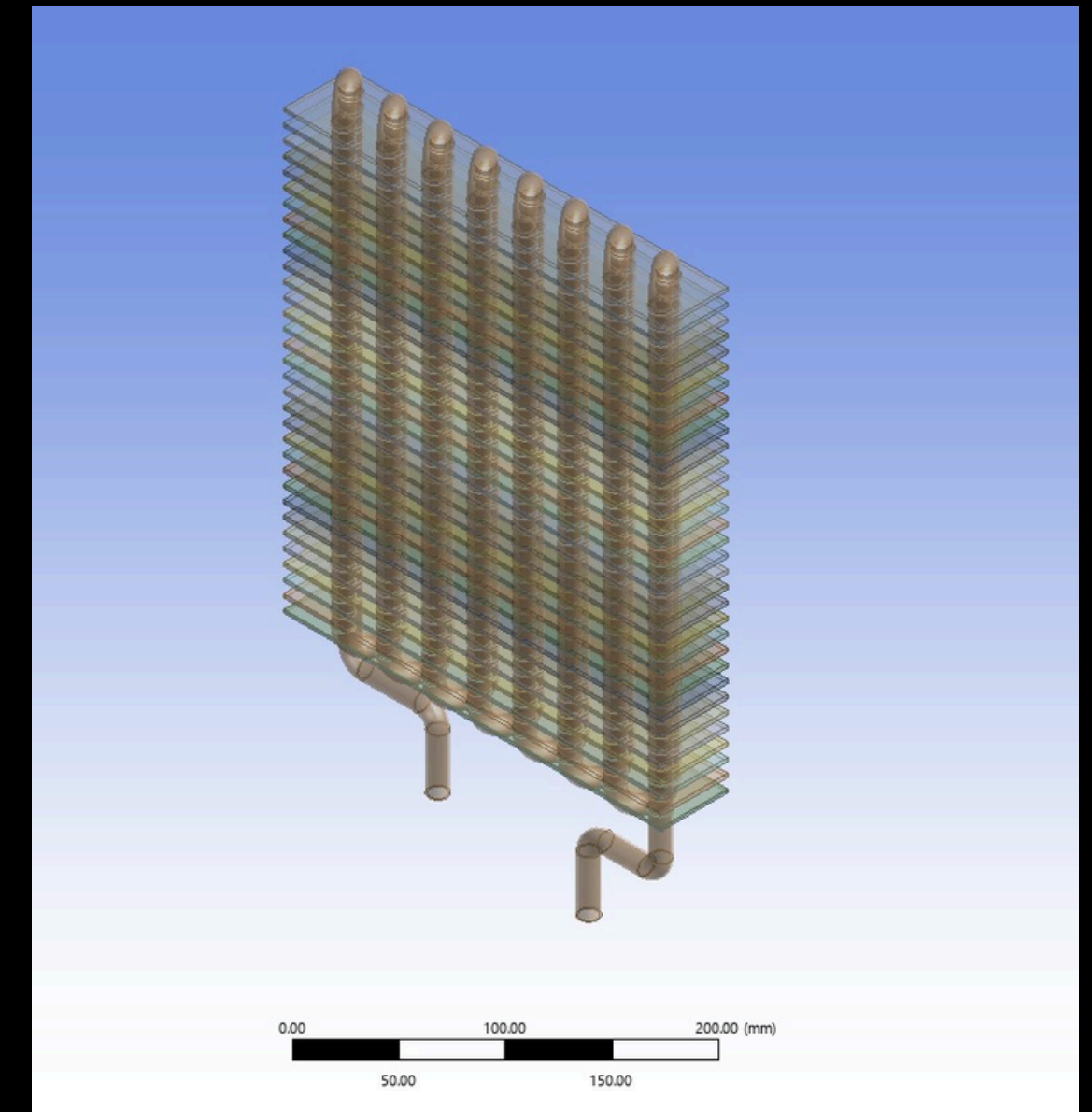
- Removed minor fillets and small features to minimize mesh size and computational cost.
- Ensured a clean and continuous interface between the fluid and solid regions.

Domain Definition:

- Fluid domain: Water (coolant)
- Solid domains: Copper (fins, pipes, and outer walls)

The finalized geometry was exported to mechanical model

Mesh for mesh generation prior to solver setup.



➤ MESHING

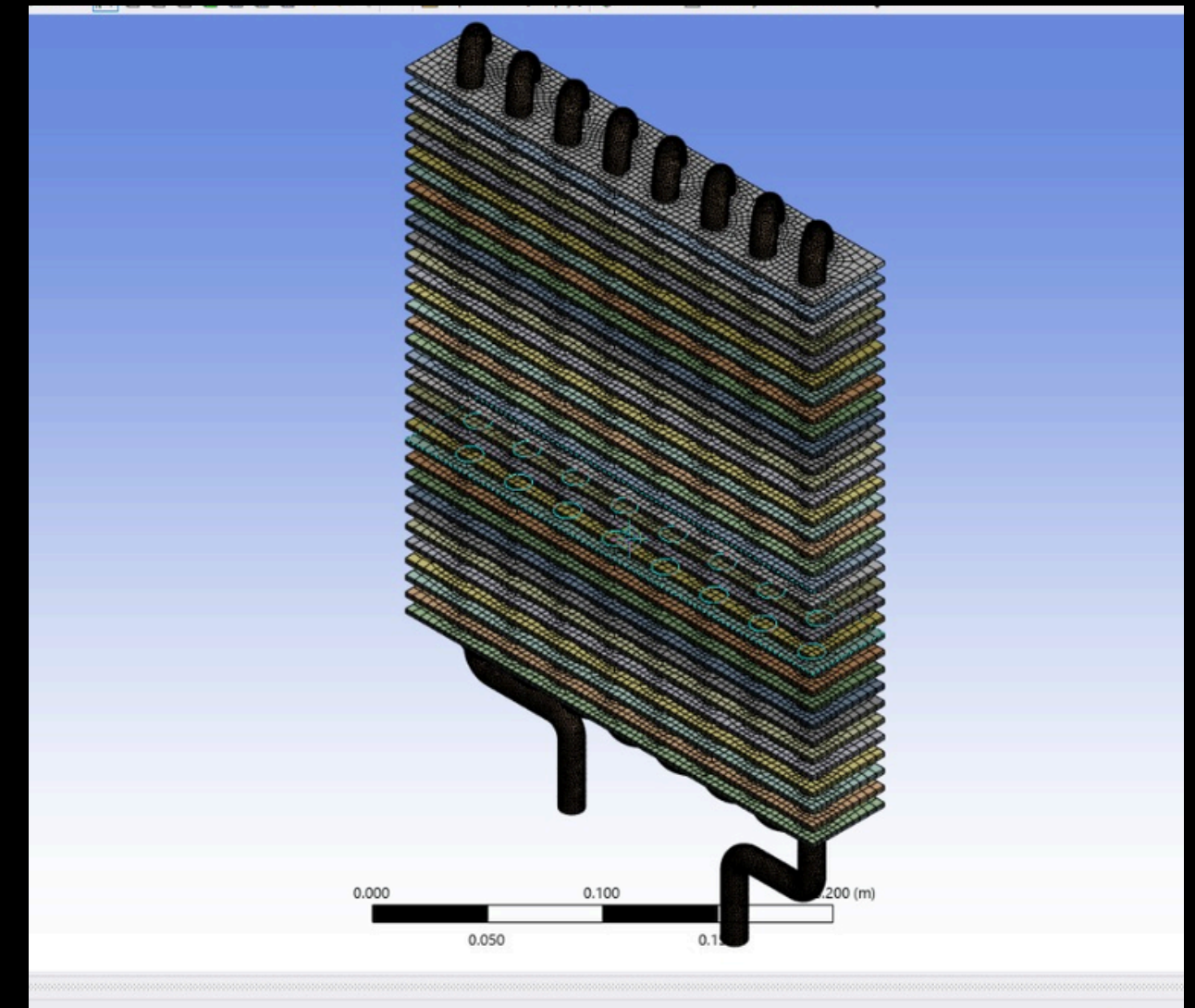


Meshing software: ANSYS Meshing mechanical model

- Meshing Approach: Unstructured tetrahedral mesh generated with inflation layers near walls.
- Element Size: Approximately 5×10^{-3} m

Quality metrics:

- Skewness < 0.9
- Orthogonal Quality > 0.2
- Boundary naming: inlet, outlet, fin-solid, pipe, walls.



➤ MESH TYPE AND METHOD

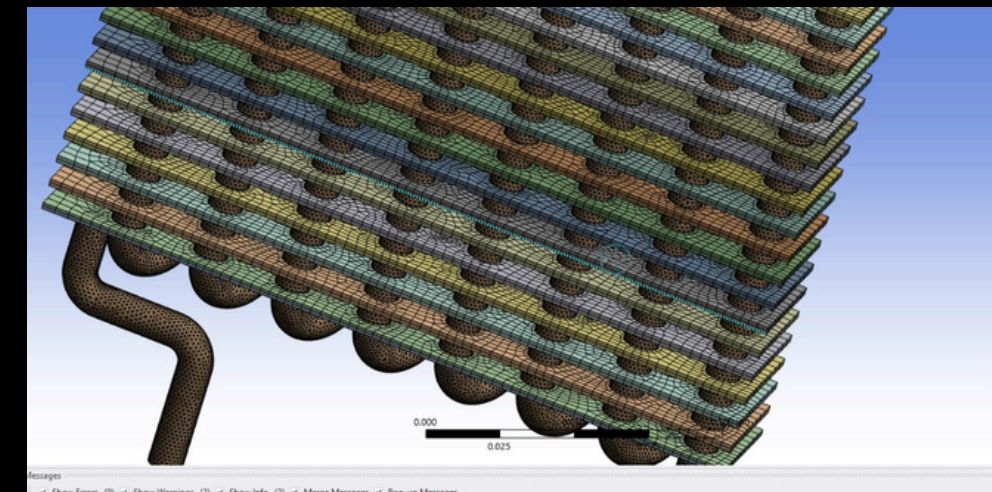
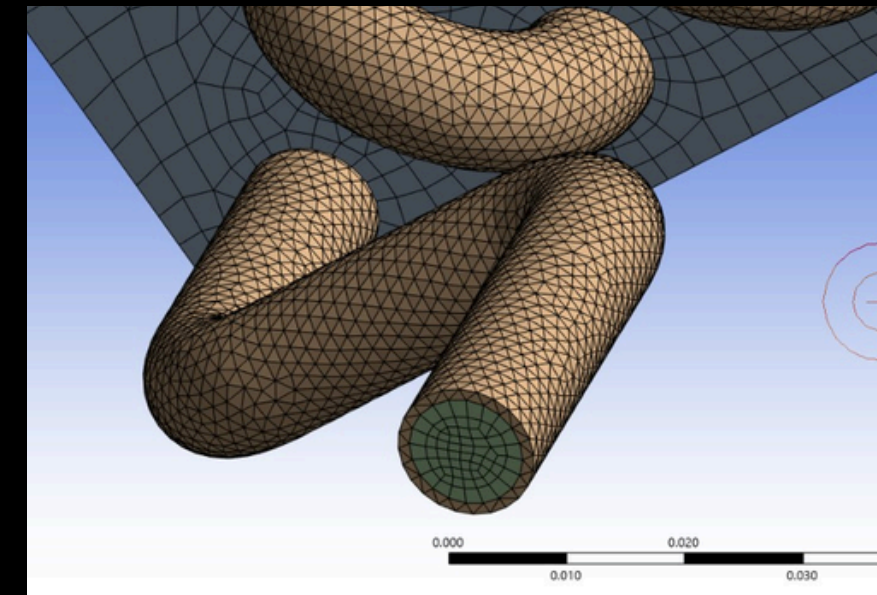
Mesh Statistics:

- Nodes: 599,111
- Elements: 851,240

Inflation Settings:

- Inflation option: Smooth transition
- Transition ratio: 0.272
- Maximum layers: 5
- Growth rate: 1.2
- Inflation algorithm: Pre

Mesh ensured acceptable element quality and smooth gradient near wall boundaries.



➤ SOLVER SET UP

Fluid :water and solid material : copper.

Solver: Pressure based, 3D, transient (unsteady)

Turbulence model: Standard k- ϵ with enhanced wall treatment

Heat transfer: Enabled

Time steps: 100 $\times$ 1 s each

Pressure velocity coupling: SIMPLE

➤ MATERIALS AND PROPERTIES

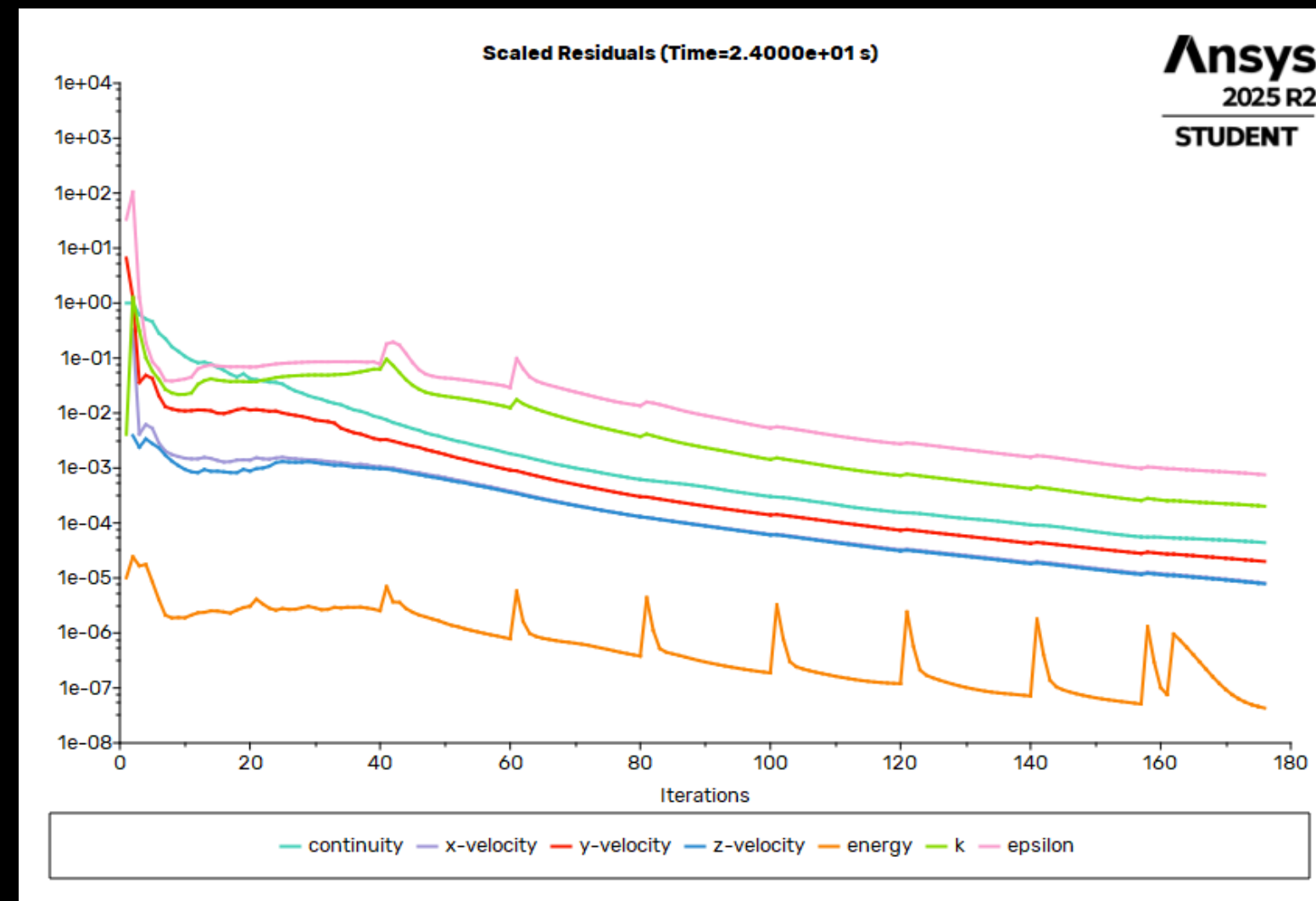
Material	Type	ρ (kg/m ³)	C_p (J/kg·K)	k (W/m·K)
Air	Fluid	1.225	1006.43	0.0242
Water	Fluid	998.2	4182	0.6
Copper	Solid	8978	381	387.6

➤ BOUNDARY CONDITIONS

- Inlet: Velocity = 0.5 m/s, Temperature = 353 K
- Outlet: Pressure outlet
- Fin & pipe walls: Convective boundary ($h = 120 \text{ W/m}^2\text{K}$)
- Outer walls: Heat flux applied (thermal load)
- All walls: Stationary, No-slip condition

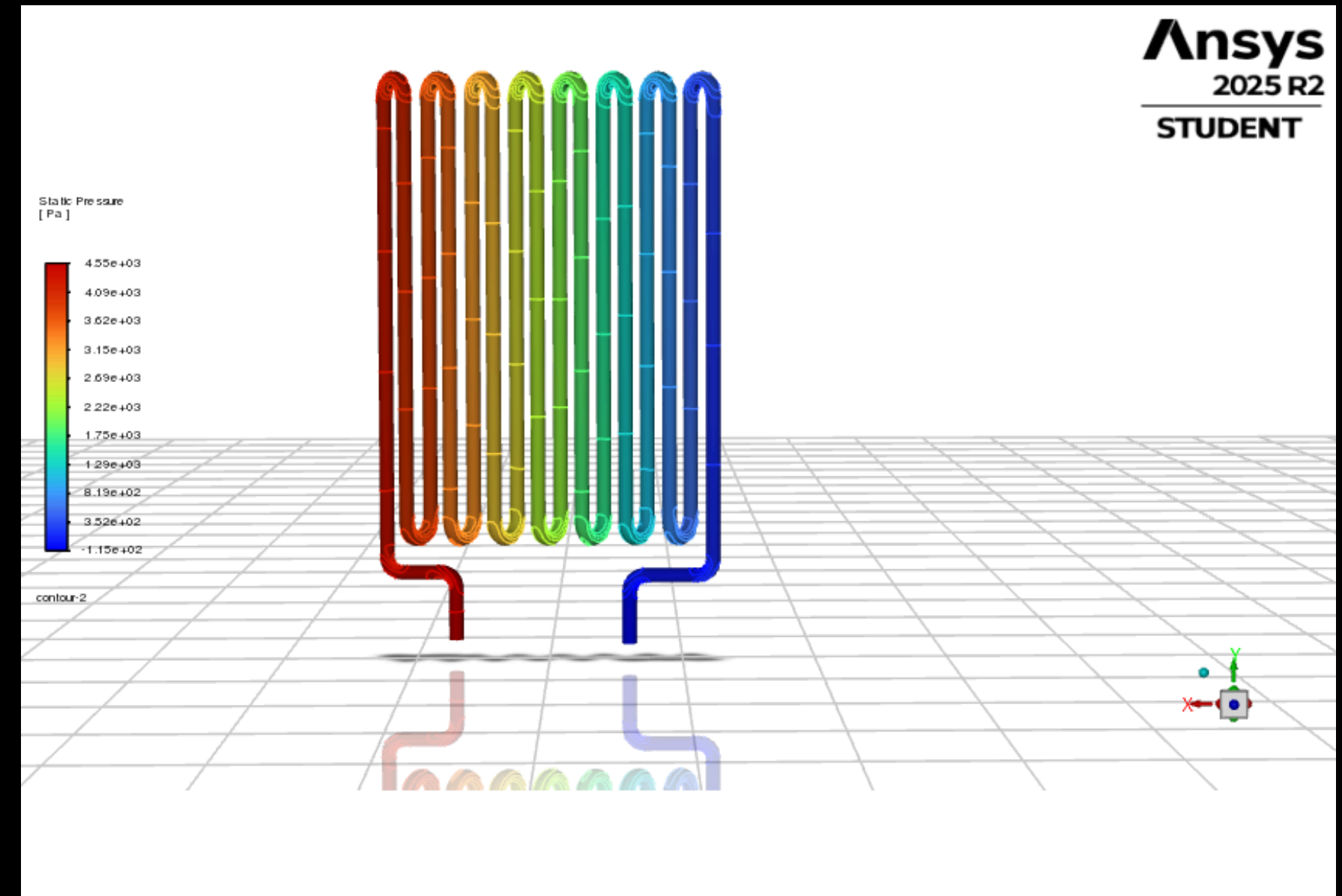
➤ RESIDUALS AND CONVERGENCE

- Monitored residuals for continuity, momentum, turbulence, and energy.
- All residuals converge below 10^{-4} , with energy residual reaching $\sim 10^{-7}$, confirming numerical stability and solution convergence.



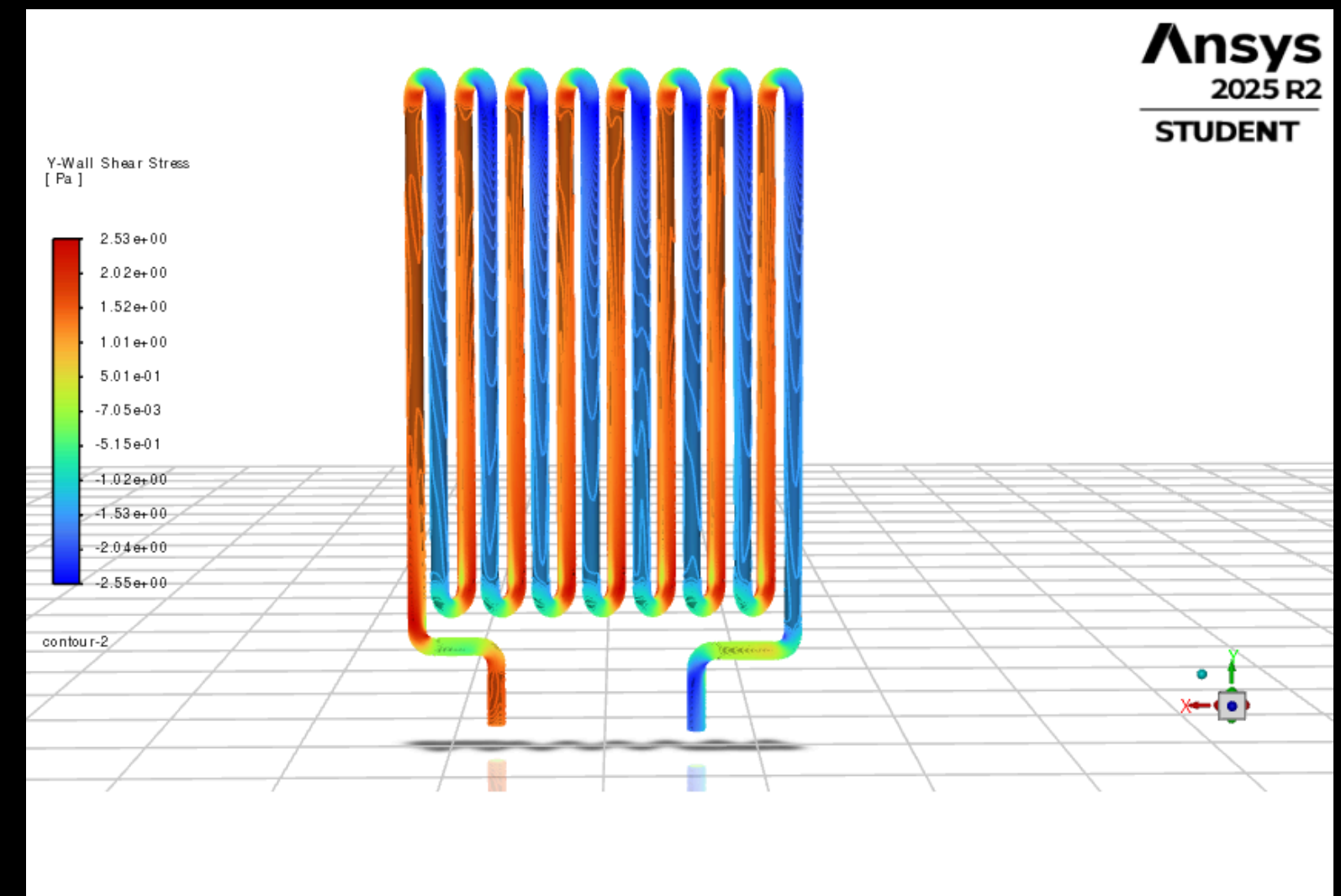
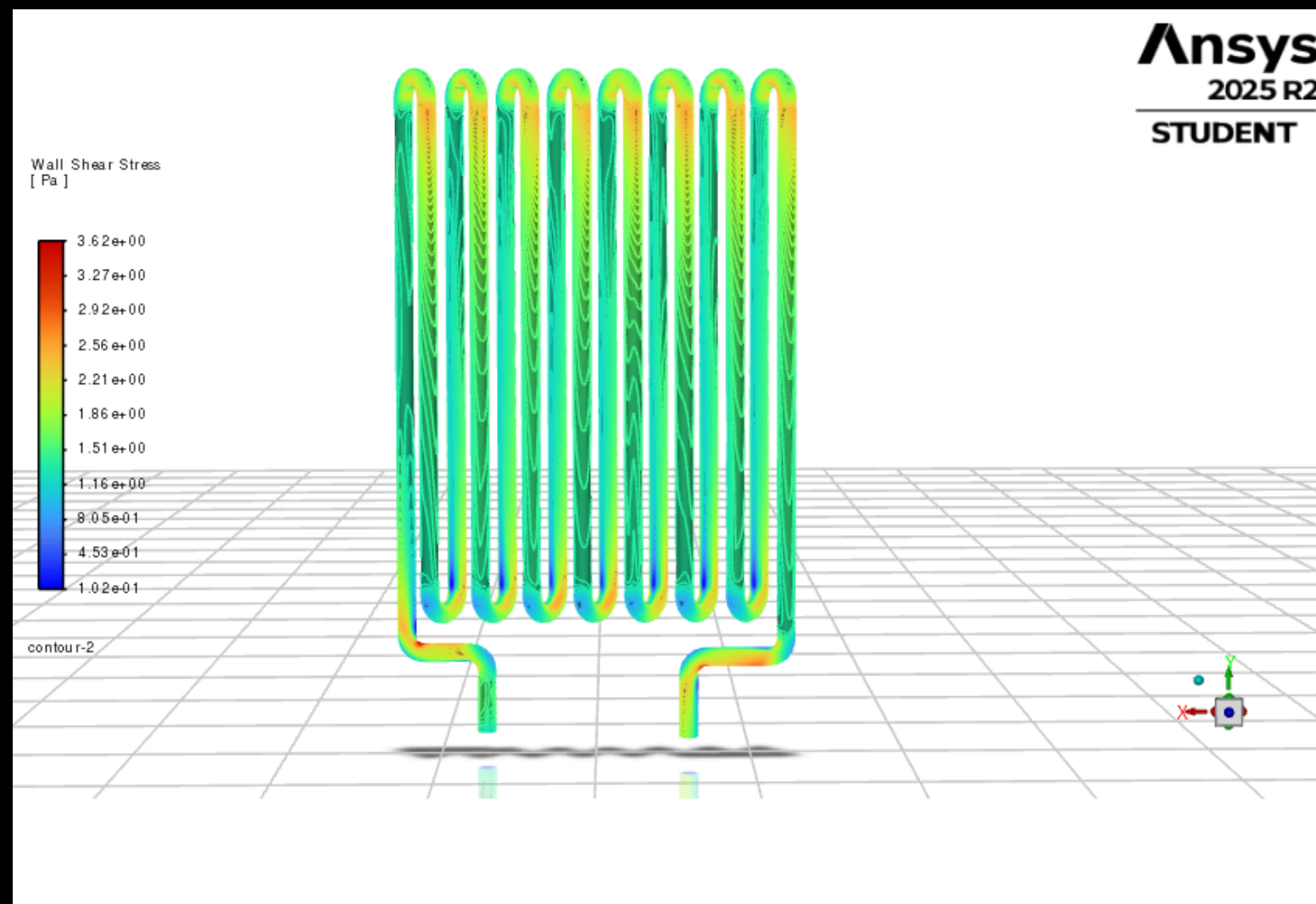
➤ PRESSURE DISTRIBUTION

- Pressure drop observed along coolant tubes.
- Indicates flow resistance and friction losses inside the radiator core.
- Pressure uniformity across outlets validates balanced tube design.



➤ VELOCITY AND SHEAR ANALYSIS

- Shear stress is highest near tube–fin junctions and wall boundaries.
- High shear → increased convective heat transfer.
- Confirms good turbulence modeling under k– ϵ regime.

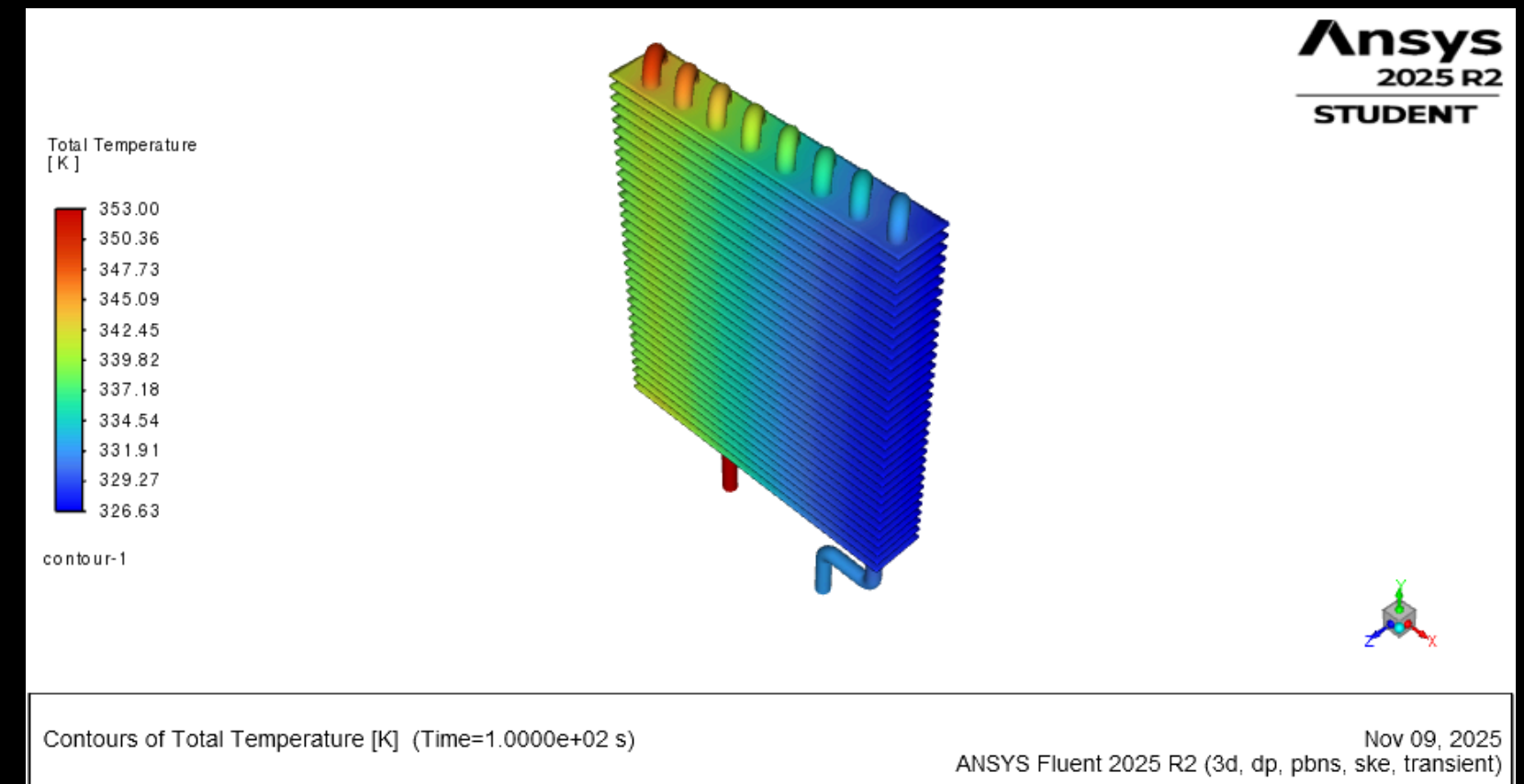
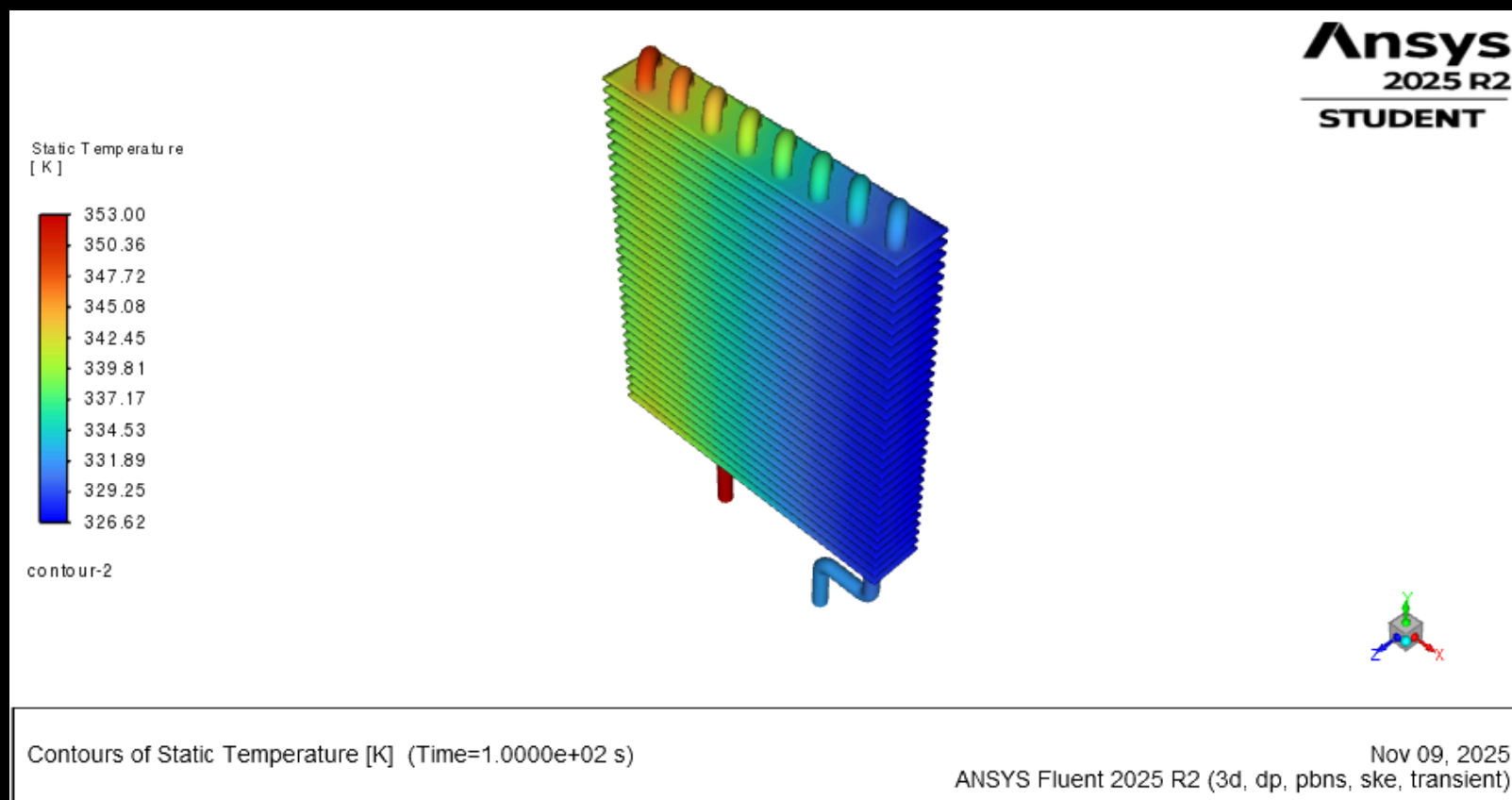


High shear stress regions indicate increased flow resistance, directly impacting pump power requirements and overall hydraulic efficiency

➤ TEMPERATURE DISTRIBUTION

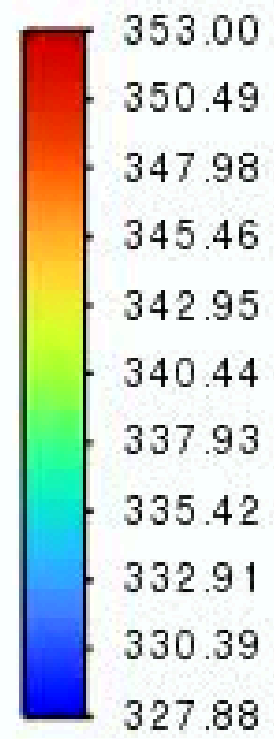
- Coolant temperature decreases gradually from inlet (353 K) to outlet.
- Fins show significant thermal gradient, indicating active heat dissipation.
- Iso-surface plots (100 s) show transient stability reached after ~80 s.

Inlet: 353 K
Outlet: ~326 K

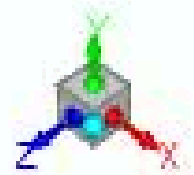


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Total Temperature
[K]



contour-1

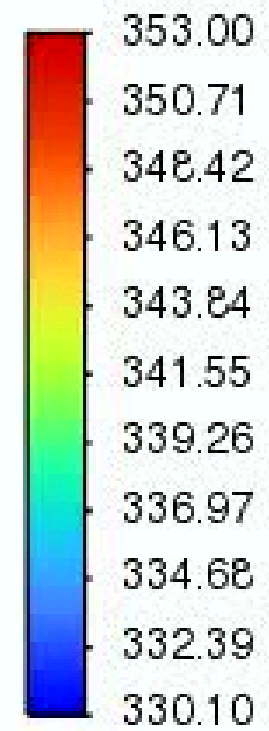


Contours of Total Temperature [K] (Time=4.5000e+01 s)

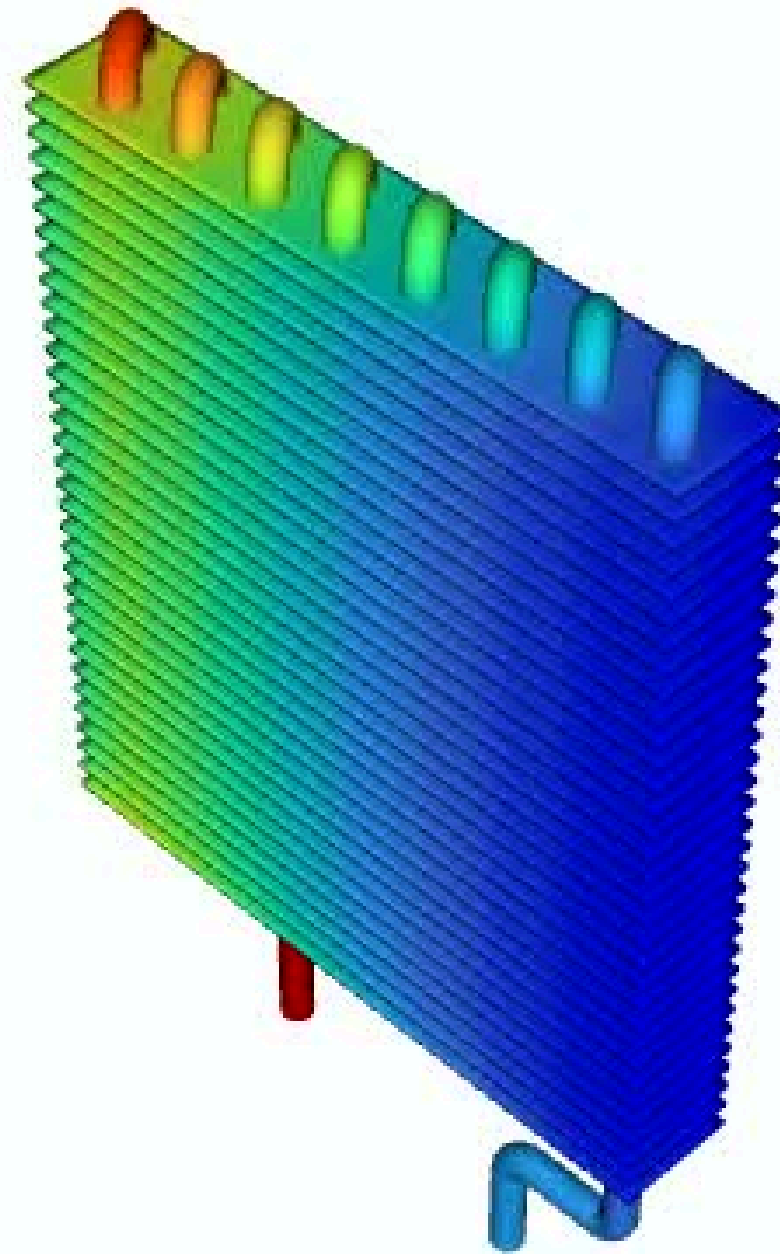
Nov 09, 2025
ANSYS Fluent 2025 R2 (3d_dp_phns_ske_transient)

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Static Temperature
[K]



contour-2



Contours of Static Temperature [K] (Time=3.2000e+01 s)

Nov 09, 2025
ANSYS Fluent 2025 R2 (3d, dp, pbns, ske, transient)

➤ KEY OBSERVATIONS

- Flow field and temperature distribution indicate effective radiator design.
- Pressure drop and heat flux values are within expected automotive range.
- Enhanced wall treatment improved near-wall temperature predictions.
- No numerical instability observed in transient regime.

CONCLUSION

- Project Goal: modeled and simulated an automobile radiator using ANSYS Fluent.
- Key Findings: The simulation showed effective fluid cooling, capturing clear temperature gradients and pressure drops.
- CFD Value: Confirmed that CFD provides valuable insight into how design changes (geometry and boundary conditions) influence performance.
- Future Improvements: While the model accurately captured essential heat transfer, future work with more detailed geometry and air-side coupling could further increase accuracy.

➤ FUTURE SCOPE

- All essential contour and graphical outputs were generated for an inlet velocity of 0.5 m/s.
- Current analysis focuses on baseline inlet velocity (0.5 m/s) to establish validation. Future phases will expand to variable flow rates.
- In the future, simulations will be extended to multiple inlet velocities to analyze performance variation.
- From these cases, we will extract:
 1. Outlet temperature for each velocity,
 2. Corresponding volumetric discharge (Q), and
 3. Mass flow rate ($\dot{m}$) calculations.
- These parameters will enable evaluation of radiator efficiency and heat transfer effectiveness under varying operating conditions.

THANK
YOU

[DRIVE LINK FOR PROJECT](#)

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